

Stochastic Subgradient Descent for Convex and Lipschitz Objectives

1 Algorithm Description

Stochastic Subgradient Descent (SSGD) is an iterative optimization algorithm designed for minimizing convex and Lipschitz continuous objective functions. The algorithm updates parameters by computing a subgradient based on a single randomly sampled data point and taking a step in the negative subgradient direction. This approach is particularly effective for large-scale problems where evaluating the full gradient is computationally expensive.

1.1 Mathematical Formulation

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a convex function satisfying Lipschitz continuity with constant L , i.e., for all $w, v \in \mathbb{R}^d$:

$$|f(w) - f(v)| \leq L\|w - v\|_2.$$

The algorithm minimizes $f(w)$ by iteratively sampling a single data point (or stochastic component) and computing a subgradient $g_t \in \partial f(w_t)$, where $\partial f(w_t)$ is the subdifferential set at w_t . The update rule is:

$$w_{t+1} = w_t - \eta_t g_t,$$

where $\eta_t > 0$ is the step size at iteration t . For Lipschitz functions, any subgradient satisfies $\|g_t\|_2 \leq L$.

1.2 Hyperparameters

1. **Initial parameters** $w_0 \in \mathbb{R}^d$: Starting point for optimization.
2. **Step size sequence** η_t : Typically chosen as $\eta_t = \frac{1}{\sqrt{t+1}}$ to ensure convergence.
3. **Number of iterations** T : Termination condition for the algorithm.

2 Pseudocode and Dry Run

2.1 Pseudocode

Algorithm 1 Stochastic Subgradient Descent (SSGD)

Initialize $w_0 \in \mathbb{R}^d$, step size schedule η_t , maximum iterations T $t = 0$ to $T - 1$
Sample a single data point $z_t \sim \mathcal{D}$ (distribution over data) Compute subgradient $g_t \in \partial f(w_t; z_t)$ Update parameters: $w_{t+1} = w_t - \eta_t g_t$ Return w_T

2.2 Dry Run Example

Consider the function $f(w) = (w - 3)^2$, with initialization $w_0 = 0$, step size $\eta = 0.1$.

Iteration 1:

- **Gradient:** $g_0 = \frac{df}{dw}(w_0) = 2(w_0 - 3) = 2(0 - 3) = -6$
- **Update:** $w_1 = 0 - 0.1 \cdot (-6) = 0 + 0.6 = 0.6$
- **Objective:** $f(w_1) = (0.6 - 3)^2 = 5.76$

Iteration 2:

- **Gradient:** $g_1 = 2(0.6 - 3) = -4.8$
- **Update:** $w_2 = 0.6 - 0.1 \cdot (-4.8) = 0.6 + 0.48 = 1.08$
- **Objective:** $f(w_2) = (1.08 - 3)^2 = 3.6864$

Iteration 3:

- **Gradient:** $g_2 = 2(1.08 - 3) = -3.84$
- **Update:** $w_3 = 1.08 - 0.1 \cdot (-3.84) = 1.08 + 0.384 = 1.464$
- **Objective:** $f(w_3) = (1.464 - 3)^2 = 2.3593$

Observations:

- The algorithm converges toward the true minimum at $w = 3$.
- Despite $f(w)$ not being globally Lipschitz continuous, the subgradient magnitude decreases as w_t approaches the minimum, enabling convergence.

3 Convergence Theory

3.1 Assumptions

Assumption 1 (Convex Objective Function). *The objective function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex. That is, for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\lambda \in [0, 1]$,*

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}).$$

Assumption 2 (Lipschitz Continuity). *There exists a constant $L > 0$ such that f is L -Lipschitz continuous:*

$$|f(\mathbf{x}) - f(\mathbf{y})| \leq L\|\mathbf{x} - \mathbf{y}\| \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n.$$

This implies that all subgradients $\mathbf{g} \in \partial f(\mathbf{x})$ satisfy $\|\mathbf{g}\| \leq L$.

Assumption 3 (Stochastic Subgradient Oracle). *At each iteration t , given a parameter vector $\mathbf{x}_t$, the algorithm receives a stochastic subgradient $\mathbf{g}_t \in \mathbb{R}^n$ satisfying:*

1. $\mathbb{E}[\mathbf{g}_t \mid \mathbf{x}_t] \in \partial f(\mathbf{x}_t)$ (unbiasedness);
2. $\mathbb{E}[\|\mathbf{g}_t\|^2 \mid \mathbf{x}_t] \leq L^2$ (bounded variance).

Assumption 4 (Bounded Domain). *The feasible set $\mathcal{X} \subset \mathbb{R}^n$ is convex and bounded with diameter $R > 0$, where:*

$$\|\mathbf{x} - \mathbf{y}\| \leq R \quad \forall \mathbf{x}, \mathbf{y} \in \mathcal{X}.$$

For unbounded domains, R denotes the distance from the initial iterate $\mathbf{x}_0$ to the optimal set $\mathcal{X}^$.*

3.2 Main Convergence Theorem

Theorem 1 (Convergence Rate of Stochastic Subgradient Descent). *Under Assumptions 1–4, let $\mathbf{x}^* \in \mathcal{X}^*$ be an optimal solution. Suppose the algorithm uses step sizes $\eta_t = \frac{R}{L\sqrt{T}}$ for T iterations. Define the averaged iterate $\bar{\mathbf{x}}_T = \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t$. Then, the expected suboptimality satisfies:*

$$\mathbb{E}[f(\bar{\mathbf{x}}_T)] - f(\mathbf{x}^*) \leq \frac{2LR}{\sqrt{T}}.$$

Proof. Using convexity and the unbiasedness of $\mathbf{g}_t$, we derive:

$$f(\mathbf{x}_t) - f(\mathbf{x}^*) \leq \mathbb{E}[\mathbf{g}_t^\top (\mathbf{x}_t - \mathbf{x}^*)].$$

By analyzing the iterates via the Pythagorean identity and telescoping sums, we bound the cumulative error:

$$\sum_{t=1}^T \mathbb{E}[f(\mathbf{x}_t) - f(\mathbf{x}^*)] \leq \frac{R^2}{2\eta_T} + \frac{L^2}{2} \sum_{t=1}^T \eta_t.$$

Substituting $\eta_t = \frac{R}{L\sqrt{T}}$ and dividing by T completes the proof. $\square$

3.3 Theoretical Properties

1. **Stability:** The iterates $\{\mathbf{x}_t\}$ remain within a ball of radius $O(R)$ centered at $\mathbf{x}_0$, ensuring numerical stability.
2. **Sensitivity to Hyperparameters:**
 - Step sizes $\eta_t = \Theta\left(\frac{1}{\sqrt{t}}\right)$ are critical for convergence. Larger steps amplify variance, while smaller steps slow progress.

- The optimal constant R/L depends on problem geometry and is typically unknown, but the rate $O(1/\sqrt{T})$ is preserved for $\eta_t \propto 1/\sqrt{t}$.

3. Comparison to Baselines:

- Compared to deterministic subgradient descent (also $O(1/\sqrt{T})$), the stochastic variant reduces per-iteration cost from $O(n)$ to $O(1)$, making it scalable for large datasets.
- For smooth objectives, gradient descent achieves $O(1/T)$ rates, highlighting the cost of nonsmoothness.

4. Special Cases:

- If f is μ -strongly convex, step sizes $\eta_t = O(1/(\mu t))$ yield $O(\log T/T)$ rates.
- Variance reduction techniques (e.g., SAGA, SVRG) can accelerate convergence under additional smoothness assumptions.

3.4 Discussion

The $O(1/\sqrt{T})$ convergence rate is tight for stochastic subgradient methods on convex Lipschitz functions, reflecting fundamental limitations of first-order optimization in noisy settings. The analysis highlights the role of domain geometry (R) and function regularity (L) in determining constants. Practical performance often benefits from adaptive stepsize rules (e.g., AdaGrad) or momentum, though theoretical guarantees for nonsmooth functions remain limited. This framework generalizes to broader settings, including online learning and robust optimization, where subgradient methods are foundational.

4 Convergence Proofs

4.1 Preliminaries (Definitions and Lemmas)

4.1.1 Definitions

Convex Function: A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\lambda \in [0, 1]$,

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}).$$

Subgradient: For a convex function f , a vector $\mathbf{g} \in \mathbb{R}^n$ is a subgradient of f at $\mathbf{x}$ if:

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \mathbf{g}^\top (\mathbf{y} - \mathbf{x}) \quad \forall \mathbf{y} \in \mathbb{R}^n.$$

The set of all subgradients at $\mathbf{x}$ is denoted $\partial f(\mathbf{x})$.

Lipschitz Continuity: A function f is L -Lipschitz continuous if:

$$|f(\mathbf{x}) - f(\mathbf{y})| \leq L \|\mathbf{x} - \mathbf{y}\| \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n.$$

This implies $\|\mathbf{g}\| \leq L$ for all $\mathbf{g} \in \partial f(\mathbf{x})$.

Projection Operator: For a convex set $\mathcal{X} \subset \mathbb{R}^n$, the projection $\Pi_{\mathcal{X}}(\mathbf{y})$ is the closest point in $\mathcal{X}$ to $\mathbf{y}$:

$$\Pi_{\mathcal{X}}(\mathbf{y}) = \arg \min_{\mathbf{x} \in \mathcal{X}} \|\mathbf{y} - \mathbf{x}\|.$$

The projection satisfies the Pythagorean identity:

$$\|\Pi_{\mathcal{X}}(\mathbf{y}) - \mathbf{x}^*\|^2 \leq \|\mathbf{y} - \mathbf{x}^*\|^2 - \|\mathbf{y} - \Pi_{\mathcal{X}}(\mathbf{y})\|^2 \quad \forall \mathbf{x}^* \in \mathcal{X}.$$

4.1.2 Lemmas

Lemma 1 (Convexity Inequality). *For any $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{g} \in \partial f(\mathbf{x})$,*

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \mathbf{g}^\top (\mathbf{y} - \mathbf{x}) \quad \forall \mathbf{y} \in \mathbb{R}^n.$$

Lemma 2 (Projection Lemma). *For any $\mathbf{y} \in \mathbb{R}^n$ and $\mathbf{x}^* \in \mathcal{X}$,*

$$\|\Pi_{\mathcal{X}}(\mathbf{y}) - \mathbf{x}^*\|^2 \leq \|\mathbf{y} - \mathbf{x}^*\|^2.$$

Lemma 3 (Jensen's Inequality). *For a convex function f and any random variable $\mathbf{x}$,*

$$f(\mathbb{E}[\mathbf{x}]) \leq \mathbb{E}[f(\mathbf{x})].$$

4.2 Main Proof (Standard Setting)

We analyze the stochastic subgradient descent (SSGD) algorithm:

$$\mathbf{x}_{t+1} = \Pi_{\mathcal{X}}(\mathbf{x}_t - \eta_t \mathbf{g}_t),$$

where $\mathbf{g}_t$ is an unbiased stochastic subgradient satisfying $\mathbb{E}[\mathbf{g}_t \mid \mathbf{x}_t] \in \partial f(\mathbf{x}_t)$ and $\mathbb{E}[\|\mathbf{g}_t\|^2 \mid \mathbf{x}_t] \leq L^2$.

4.2.1 Step 1: Distance Recursion

Using the projection lemma:

$$\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 \leq \|\mathbf{x}_t - \eta_t \mathbf{g}_t - \mathbf{x}^*\|^2.$$

Expanding the right-hand side:

$$= \|\mathbf{x}_t - \mathbf{x}^*\|^2 - 2\eta_t \mathbf{g}_t^\top (\mathbf{x}_t - \mathbf{x}^*) + \eta_t^2 \|\mathbf{g}_t\|^2.$$

Taking expectations conditioned on $\mathbf{x}_t$:

$$\mathbb{E}[\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 \mid \mathbf{x}_t] \leq \|\mathbf{x}_t - \mathbf{x}^*\|^2 - 2\eta_t \mathbb{E}[\mathbf{g}_t^\top (\mathbf{x}_t - \mathbf{x}^*) \mid \mathbf{x}_t] + \eta_t^2 \mathbb{E}[\|\mathbf{g}_t\|^2 \mid \mathbf{x}_t].$$

4.2.2 Step 2: Subgradient Inequality

From convexity (Lemma 1):

$$\mathbb{E}[\mathbf{g}_t^\top (\mathbf{x}_t - \mathbf{x}^*) \mid \mathbf{x}_t] \geq \mathbb{E}[f(\mathbf{x}_t) - f(\mathbf{x}^*) \mid \mathbf{x}_t].$$

Using the tower property:

$$\mathbb{E}[\mathbf{g}_t^\top (\mathbf{x}_t - \mathbf{x}^*)] \geq \mathbb{E}[f(\mathbf{x}_t) - f(\mathbf{x}^*)].$$

Also, $\mathbb{E}[\|\mathbf{g}_t\|^2] \leq L^2$. Substituting:

$$\mathbb{E}[\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2] \leq \mathbb{E}[\|\mathbf{x}_t - \mathbf{x}^*\|^2] - 2\eta_t \mathbb{E}[f(\mathbf{x}_t) - f(\mathbf{x}^*)] + \eta_t^2 L^2.$$

4.2.3 Step 3: Telescoping Sum

Summing over $t = 0$ to $T - 1$:

$$\sum_{t=0}^{T-1} 2\eta_t \mathbb{E}[f(\mathbf{x}_t) - f(\mathbf{x}^*)] \leq \mathbb{E}[\|\mathbf{x}_0 - \mathbf{x}^*\|^2] - \mathbb{E}[\|\mathbf{x}_T - \mathbf{x}^*\|^2] + \sum_{t=0}^{T-1} \eta_t^2 L^2.$$

Since $\|\mathbf{x}_0 - \mathbf{x}^*\| \leq R$, we have $\mathbb{E}[\|\mathbf{x}_0 - \mathbf{x}^*\|^2] \leq R^2$. Dropping the non-negative term $\mathbb{E}[\|\mathbf{x}_T - \mathbf{x}^*\|^2]$:

$$\sum_{t=0}^{T-1} 2\eta_t \mathbb{E}[f(\mathbf{x}_t) - f(\mathbf{x}^*)] \leq R^2 + \sum_{t=0}^{T-1} \eta_t^2 L^2.$$

4.2.4 Step 4: Averaged Iterate Bound

Assuming constant step size $\eta_t = \eta$, dividing both sides by $2\eta T$:

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[f(\mathbf{x}_t) - f(\mathbf{x}^*)] \leq \frac{R^2}{2\eta T} + \frac{\eta L^2}{2}.$$

By Jensen's inequality (Lemma 3):

$$\mathbb{E}[f(\bar{\mathbf{x}}_T)] - f(\mathbf{x}^*) \leq \frac{R^2}{2\eta T} + \frac{\eta L^2}{2},$$

where $\bar{\mathbf{x}}_T = \frac{1}{T} \sum_{t=0}^{T-1} \mathbf{x}_t$.

4.2.5 Rate Derivation

Substitute $\eta = \frac{R}{L\sqrt{T}}$:

1. **First Term:**

$$\frac{R^2}{2\eta T} = \frac{R^2}{2 \cdot \frac{R}{L\sqrt{T}} \cdot T} = \frac{RL}{2\sqrt{T}}.$$

2. **Second Term:**

$$\frac{\eta L^2}{2} = \frac{\frac{R}{L\sqrt{T}} \cdot L^2}{2} = \frac{RL}{2\sqrt{T}}.$$

Adding both terms:

$$\mathbb{E}[f(\bar{\mathbf{x}}_T)] - f(\mathbf{x}^*) \leq \frac{RL}{2\sqrt{T}} + \frac{RL}{2\sqrt{T}} = \frac{RL}{\sqrt{T}}.$$

4.3 Proof under Strongly Convex Objective

If f is μ -strongly convex:

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \mathbf{g}^\top (\mathbf{y} - \mathbf{x}) + \frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|^2.$$

Using step sizes $\eta_t = \frac{c}{\mu t}$, we follow a similar derivation. The key difference is the strong convexity term, which provides a quadratic lower bound, improving the rate. Summing the recursion and optimizing the step size yields:

$$\mathbb{E}[f(\bar{\mathbf{x}}_T)] - f(\mathbf{x}^*) \leq \frac{C \log T}{T},$$

where C depends on L , μ , and R .

The proof follows the same steps as above, with the subgradient inequality modified to include the strong convexity term. The telescoping sum incorporates the quadratic term, leading to a logarithmic factor in the numerator.

4.4 Proof with Diminishing Step Sizes

Using step sizes $\eta_t = \frac{R}{L\sqrt{t+1}}$, the proof adapts the telescoping sum:

$$\sum_{t=0}^{T-1} \eta_t^2 = \sum_{t=0}^{T-1} \frac{R^2}{L^2(t+1)} \approx \frac{R^2}{L^2} \log T.$$

The suboptimality bound becomes:

$$\mathbb{E}[f(\bar{\mathbf{x}}_T)] - f(\mathbf{x}^*) \leq O\left(\frac{\log T}{\sqrt{T}}\right).$$

This proof follows the same structure, adjusting the summation terms to account for the diminishing steps.

4.5 Proof with Bounded Subgradient Variance

Assuming $\mathbb{E}[\|\mathbf{g}_t\|^2 \mid \mathbf{x}_t] \leq \sigma^2$ (instead of L^2), the proof adjusts the variance term:

$$\mathbb{E}[\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2] \leq \mathbb{E}[\|\mathbf{x}_t - \mathbf{x}^*\|^2] - 2\eta_t \mathbb{E}[f(\mathbf{x}_t) - f(\mathbf{x}^*)] + \eta_t^2 \sigma^2.$$

Using $\eta = \frac{R}{\sigma\sqrt{T}}$, the rate becomes:

$$\mathbb{E}[f(\bar{\mathbf{x}}_T)] - f(\mathbf{x}^*) \leq \frac{R\sigma}{\sqrt{T}}.$$

The structure remains identical, with σ replacing L in the variance bound.

4.6 Proof for Unbounded Domain

For unbounded domains, assume $\|\mathbf{x}_0 - \mathbf{x}^*\| \leq R_0$. The proof structure is unchanged, but R_0 replaces R . The rate remains:

$$\mathbb{E}[f(\bar{\mathbf{x}}_T)] - f(\mathbf{x}^*) \leq \frac{R_0 L}{\sqrt{T}}.$$

The projection step may be omitted if the domain is unconstrained, simplifying the update to $\mathbf{x}_t = 0 - \mathbf{x}_{t+1} = \eta_t \mathbf{g}_t$.

4.7 Proof with Adaptive Step Sizes

Using adaptive step sizes (e.g., AdaGrad), the step size is scaled by the cumulative norm of subgradients. The proof requires additional analysis of the adaptive terms, but the rate remains $O(1/\sqrt{T})$ under standard assumptions. The derivation follows the same steps, with η_t replaced by the adaptive rule, and the summation bounds are adjusted accordingly.

4.8 Conclusion

Under the given assumptions, the stochastic subgradient descent algorithm with constant step size $\eta_t = \frac{R}{L\sqrt{T}}$ achieves an expected suboptimality bound of $\frac{RL}{\sqrt{T}}$ for the averaged iterate. Alternative settings (strong convexity, diminishing steps, etc.) modify the constants or logarithmic factors but preserve the rate. These results highlight the robustness of the SSGD across various conditions.